VESIT Strategic SDG Alignment and Action Framework 2024-2030

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1. INSTITUTIONAL STATEMENT OF PURPOSE

Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai, adopts the United Nations 2030 Agenda for Sustainable Development as the foundational guiding framework for its capital infrastructure, educational curricula, and research operations. As an institution of 3,200 individuals operating across 32,500 m2 of educational infrastructure, VESIT recognizes that higher education institutions possess an ethical obligation to demonstrate evidence-based sustainability. While affirming all 17 Sustainable Development Goals, this strategic framework concentrates institutional resources and research capabilities on seven core SDGs where campus operations exert direct, quantifiable impact.

2. MAPPING INSTITUTIONAL INITIATIVES AND TARGETS TO UN SDGS

2.1 SDG 6: Clean Water and Sanitation:

- Key Campus Initiative: Integrated Rainwater Harvesting & Groundwater Recharge Cisterns (INIT_RWH_SYSTEM).

- Core Campus Target: 30% Reduction in Municipal Water Consumption by 2029 (TRG_WATER_NEUTRAL).

- Measurable Metrics: Captures 12,400,000 litres of monsoon rainwater annually from 18,000 m2 terrace catchments across Academic Block A (BLD_ACA), Block B (BLD_ACB), and Library (BLD_LIB). Eliminates 25.8% of municipal water demand and provides zero untreated effluent discharge to municipal sewers.

2.2 SDG 7: Affordable and Clean Energy:

- Key Campus Initiatives: Phase-1 Rooftop Solar PV Project (INIT_SOLAR_PH1, 220 kWp) and Phase-2 Solar Canopy and Storage Proposal (INIT_SOLAR_PH2, 200 kWp + 100 kWh BESS).

- Core Campus Target: 50% Renewable Electricity Generation by 2030 (TRG_RENEWABLE).

- Measurable Metrics: 312,400 kWh generated annually in Phase 1, expanding to 596,400 kWh in Phase 2. Offsets fossil grid electricity (EF_ELEC_GRID = 0.82 kg CO2e/kWh), avoiding over 489,000 kg CO2e annually.

2.3 SDG 9: Industry, Innovation and Infrastructure:

- Key Campus Initiative: Campus-Wide IoT Smart Sub-Metering & Energy Audit Protocol (INIT_IOT_ENERGY_MGT).

- Measurable Metrics: 45 Modbus-RS485 digital smart meters, edge computing gateways, and AI telemetry developed by Department of Computer Engineering (DEP_CMPN) and AI & Data Science (DEP_AIDS), cutting campus standby consumption by 98,000 kWh/year.

2.4 SDG 11: Sustainable Cities and Communities:

- Key Campus Initiative: Zero-Emission Electric Campus Transit & Station Feeder Van (INIT_ELECTRIC_SHUTTLE).

- Core Campus Target: 70% Low-Carbon Campus Commute Share by 2027 (TRG_GREEN_COMMUTE).

- Measurable Metrics: Two electric vans delivering 38,400 clean passenger-trips between Kurla/Chembur stations and campus, avoiding 36,000 kg CO2e of fossil fuel commuting emissions.

2.5 SDG 12: Responsible Consumption and Production:

- Key Campus Initiative: Smart Food Waste Digester & Organic Aerobic Composting Pit (INIT_BIOGAS_CANTEEN) and E-Waste Disposal Protocol.

- Core Campus Target: Zero Waste to Landfill & 85% Recycling by 2028 (TRG_ZERO_WASTE).

- Measurable Metrics: 19,200 kg of canteen organic food waste processed into cooking biogas annually; 19,800 kg paper recycled; 1,850 kg electronic hardware recycled with Greenscape Solutions under E-Waste Rules 2022.

2.6 SDG 13: Climate Action:

- Core Campus Target: Campus Net-Zero Carbon Neutrality by 2040 (TRG_NET_ZERO).

- Measurable Metrics: Abating institutional baseline emissions of 2,450,000 kg CO2e/year along an annual linear reduction schedule of 153,125 kg CO2e/year, verified using the EcoClock AI GraphRAG analytics engine.

2.7 SDG 15: Life on Land:

- Key Campus Initiative: Miyawaki Urban Native Afforestation & Campus Biodiversity Arboretum (INIT_BIODIVERSITY_ZONE).

- Measurable Metrics: 1,200 native trees and shrubs across 32 species planted on 600 m2 behind the Auditorium (BLD_AUD), sequestering 14,000 kg CO2e annually and mitigating urban heat island temperatures by 2.3°C.

3. GOVERNANCE, MONITORING, AND ANNUAL AUDIT PROTOCOL

The Institutional Sustainability Directorate shall publish an Annual SDG Progress Scorecard every December. All metrics shall be verified through graph-grounded provenance pipelines to prevent academic greenwashing and ensure strict alignment with international sustainability standards.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_SDG_009**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*